

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

AI-Based Smart Document Summarizer

Team Members

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1. INTRODUCTION

1.1 Purpose

The purpose of this Software Requirements Specification is to define the functional and non-functional requirements of the **AI-Based Smart Document Summarizer**.

The proposed system is designed to automatically analyze lengthy documents and generate concise, meaningful, and context-aware summaries using Artificial Intelligence, Natural Language Processing (NLP), and Large Language Models (LLMs).

The system will help students, researchers, professionals, and general users quickly understand large documents without reading the entire content manually.

1.2 Project Scope

The system will allow users to upload documents such as PDF, DOCX, and TXT files. The application will extract textual information, process the content using AI models, and generate summaries based on the user's selected preferences.

The system may also provide:

- Key-point extraction
- Important keyword identification
- Multiple summary lengths
- Document-based question answering
- Summary history
- Downloadable summaries

1.3 Intended Users

The intended users include:

- Students
- Researchers
- Teachers and academicians
- Working professionals
- Organizations
- General users dealing with lengthy documents

1.4 Definitions and Abbreviations

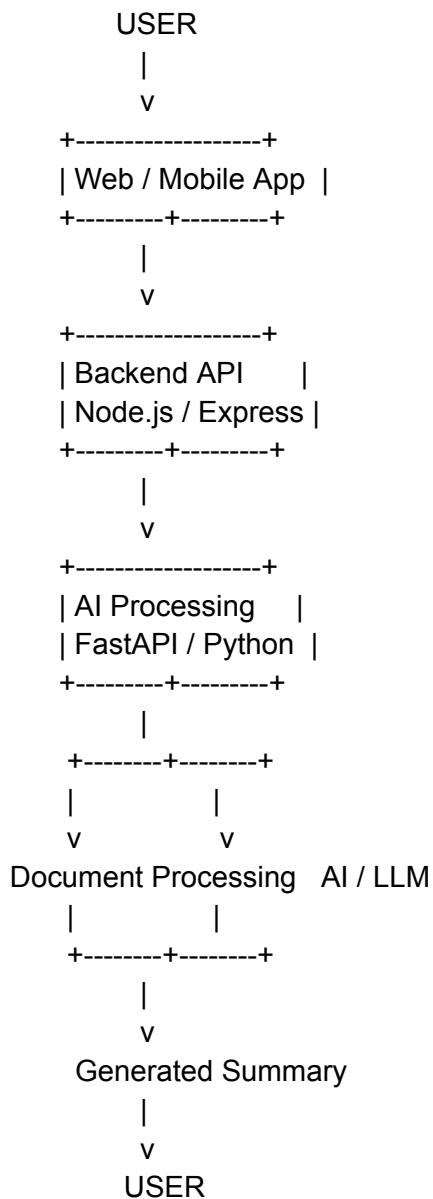
Term	Description
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model
API	Application Programming Interface
PDF	Portable Document Format
DOCX	Microsoft Word Document
RAG	Retrieval-Augmented Generation
UI	User Interface
JWT	JSON Web Token
DB	Database

2. OVERALL DESCRIPTION

2.1 Product Perspective

The AI-Based Smart Document Summarizer will be a software application consisting of a frontend interface, backend services, document-processing modules, AI processing services, and a database.

The system will follow this general architecture:



2.2 Product Functions

The major functions of the system include:

1. User registration and authentication
2. Document upload
3. Document validation
4. Text extraction
5. Text preprocessing
6. Document chunking
7. AI-based summarization
8. Key-point extraction
9. Keyword identification
10. Summary customization
11. Document-based question answering
12. Summary storage
13. Summary download
14. User history management

2.3 User Classes

Normal User

A normal user can:

- Register and log in
- Upload documents
- Generate summaries
- Ask questions about documents
- View previous summaries
- Download summaries

Administrator

An administrator, if implemented, can:

- Manage users
- Monitor system usage
- Manage uploaded content
- Monitor system performance
- Handle inappropriate or problematic content

2.4 Operating Environment

The system should operate on:

Web

- Google Chrome
- Microsoft Edge
- Mozilla Firefox
- Safari

Mobile

- Android
- iOS

Server

- Linux/Windows server
- Cloud-based deployment
- Docker-supported environment

2.5 Design Constraints

The system may have the following constraints:

- Maximum document size may be limited.
- AI model context windows may limit document processing.
- API-based LLMs may have rate limits.
- Processing time depends on document size.
- Internet connectivity may be required for cloud-based AI models.
- AI-generated summaries may occasionally contain inaccuracies.

2.6 Assumptions and Dependencies

The system assumes that:

- Users provide valid and readable documents.
 - Uploaded documents contain extractable text.
 - Required AI services are available.
 - Internet connectivity is available when cloud APIs are used.
 - The backend and database services are operational.
-

3. FUNCTIONAL REQUIREMENTS

FR-01: User Registration

The system shall allow new users to create an account using required credentials.

Requirements

- User shall provide required registration information.
 - System shall validate the input.
 - System shall store user information securely.
 - Duplicate accounts should be prevented.
-

FR-02: User Authentication

The system shall allow registered users to securely log in.

Requirements

- System shall verify user credentials.
 - Authentication shall be handled securely.
 - User sessions shall be maintained.
 - Unauthorized users shall not access protected resources.
-

FR-03: Document Upload

The system shall allow users to upload supported documents.

Supported formats

- PDF
- DOCX
- TXT

The system shall validate:

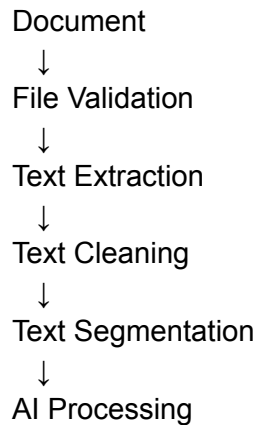
- File type
- File size

- File readability
-

FR-04: Document Processing

The system shall extract textual content from uploaded documents.

The processing pipeline shall include:



FR-05: Text Preprocessing

The system shall preprocess extracted text before sending it to the AI model.

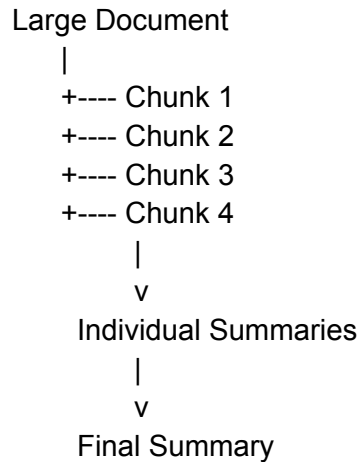
Preprocessing may include:

- Removing unnecessary whitespace
 - Removing unwanted characters
 - Normalizing text
 - Identifying sections
 - Removing redundant content
-

FR-06: Document Chunking

For large documents, the system shall divide the document into smaller chunks.

This will allow the AI model to process documents that exceed its context limitations.



FR-07: AI-Based Summarization

The system shall generate a summary of the uploaded document using an AI/LLM model.

The generated summary should:

- Represent the main ideas
 - Preserve important information
 - Avoid unnecessary repetition
 - Maintain document context
 - Be understandable to the user
-

FR-08: Summary Length Selection

The system shall allow users to select the desired summary length.

Possible options:

- Short
 - Medium
 - Detailed
-

FR-09: Key Point Extraction

The system shall identify and display important points from the document.

Example:

Key Points

- Main objective
 - Major findings
 - Important observations
 - Conclusions
 - Recommendations
-

FR-10: Keyword Extraction

The system should identify important keywords or concepts from the document.

These keywords can help users understand the main topics covered.

FR-11: Document Question Answering

The system shall optionally allow users to ask questions related to an uploaded document.

Example:

Question: What is the main objective of the research?

System: The main objective of the research is...

This functionality can be implemented using Retrieval-Augmented Generation (RAG).

FR-12: Summary History

The system shall maintain a history of previously generated summaries for authenticated users.

Users should be able to:

- View previous documents
- View generated summaries

- Delete history
 - Reopen previous results
-

FR-13: Download Summary

The system shall allow users to download generated summaries.

Possible formats:

- TXT
 - PDF
 - DOCX
-

FR-14: Error Handling

The system shall display appropriate error messages when:

- Unsupported files are uploaded.
 - File extraction fails.
 - AI service is unavailable.
 - The document is empty.
 - The document exceeds the maximum size.
 - Server errors occur.
-

4. NON-FUNCTIONAL REQUIREMENTS

4.1 Performance

- The system should process normal-sized documents within an acceptable time.
- API response times should be minimized.
- Large documents should be processed using efficient chunking.
- The system should support multiple users simultaneously.

4.2 Scalability

The system should be designed so that additional users and documents can be supported without major architectural changes.

The backend should support:

- Horizontal scaling
- Independent AI services
- Cloud deployment
- Database scaling

4.3 Security

The system shall:

- Secure user authentication.
- Encrypt sensitive information where appropriate.
- Prevent unauthorized document access.
- Validate uploaded files.
- Protect APIs against unauthorized requests.
- Use secure authentication mechanisms such as JWT or equivalent.

4.4 Reliability

The system should:

- Handle unexpected errors gracefully.
- Prevent data loss.
- Provide meaningful error messages.
- Maintain service availability.
- Recover from temporary API failures where possible.

4.5 Usability

The interface should be:

- Simple
- Responsive
- Intuitive
- Accessible to non-technical users

A user should be able to upload a document and generate a summary with minimal steps.

4.6 Maintainability

The software should use modular architecture so that individual components can be modified independently.

For example:

Frontend
Backend
AI Service
Document Processor
Database
Authentication

Each component should have clearly defined responsibilities.

4.7 Compatibility

The system should work across:

- Desktop browsers
- Mobile browsers
- Android devices
- iOS devices

4.8 Accuracy

The generated summary should accurately represent the original document.

The system should minimize:

- Hallucinations
- Missing key information
- Repetition
- Incorrect interpretation

5. EXTERNAL INTERFACE REQUIREMENTS

5.1 User Interface

The system shall provide a graphical user interface for:

- Login/registration
- Document upload
- Summary configuration
- Summary viewing
- Question answering
- Summary download
- History management

Main Dashboard

```
+-----+
|  AI SMART DOCUMENT SUMMARIZER  |
+-----+
|                                |
| Upload Document                |
| [ Choose File ]                |
|                                |
| Summary Length                 |
| ( ) Short ( ) Medium ( ) Detailed |
|                                |
|      [ SUMMARIZE ]              |
|                                |
+-----+
```

5.2 Backend API Interface

The frontend shall communicate with the backend through REST APIs.

Example endpoints:

POST /api/auth/register

POST /api/auth/login

POST /api/documents/upload

GET /api/documents

GET /api/documents/{id}

POST /api/summarize

GET /api/summaries/{id}

POST /api/documents/{id}/ask

DELETE /api/documents/{id}

5.3 AI Service Interface

The backend shall communicate with the AI service through APIs.

The AI service will receive:

Document Text

+

Summary Configuration

+

User Query (if applicable)

and return:

Generated Summary

+

Key Points

+

Keywords

5.4 Database Interface

The application shall communicate with the database to store:

- User information
- Authentication data
- Document metadata
- Summary information
- User history
- Application configuration

MongoDB can be used as the primary database.

5.5 Mobile Application Interface

The mobile application shall communicate with the same backend APIs used by the web application.

Possible technology:

React Native

This allows the team to maintain a common backend while providing both web and mobile experiences.

6. OTHER REQUIREMENTS

6.1 Data Requirements

The system may store:

User Data

- User ID
- Name
- Email
- Authentication information

Document Data

- Document ID
- File name
- File type
- Upload date
- User ID
- Document status

Summary Data

- Summary ID
- Document ID
- Summary text
- Summary type
- Generated date

6.2 Privacy Requirements

Uploaded documents may contain confidential information. Therefore:

- Documents should only be accessible to authorized users.

- User data should not be exposed to other users.
- Temporary processing files should be deleted when no longer required.
- Third-party AI APIs should be used according to their applicable data-handling policies.

6.3 Backup Requirements

Important user and application data should be backed up periodically to prevent permanent data loss.

6.4 Logging Requirements

The system should maintain logs for:

- API errors
- Authentication failures
- Document-processing failures
- AI service failures
- System errors

Logs should not unnecessarily contain sensitive document content.

6.5 Deployment Requirements

The system should be deployable using:

- Docker
- Cloud hosting
- CI/CD pipeline

The application should have separate environments for:

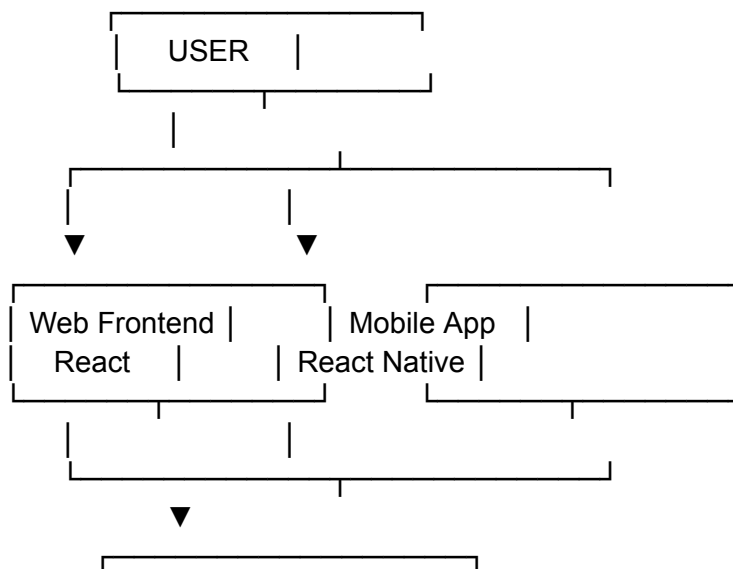
- Development
- Testing
- Production

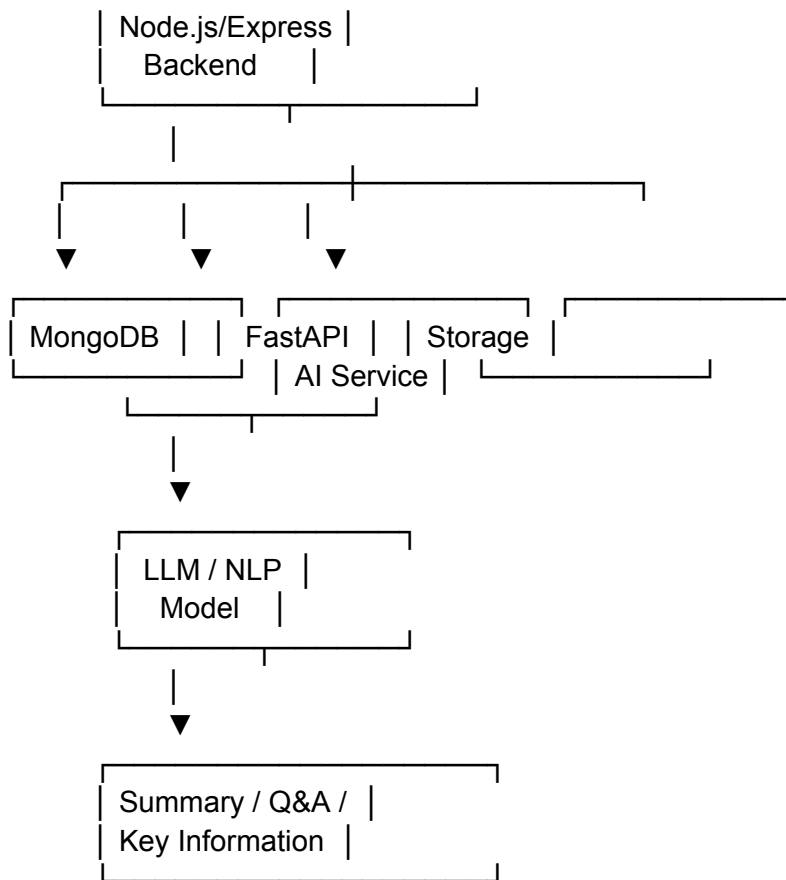
7. APPENDICES

Appendix A – Proposed Technology Stack

Layer	Technology
Web Frontend	React.js
Mobile Application	React Native
MERN Backend	Node.js + Express.js
AI Backend	Python + FastAPI
AI/LLM	Gemini / OpenAI / Open-source LLM
NLP	Transformers / Sentence Transformers
Database	MongoDB
Vector Database	FAISS / ChromaDB
PDF Processing	PyMuPDF
DOCX Processing	python-docx
Authentication	JWT
Containerization	Docker
Version Control	Git & GitHub

Appendix B – High-Level System Architecture



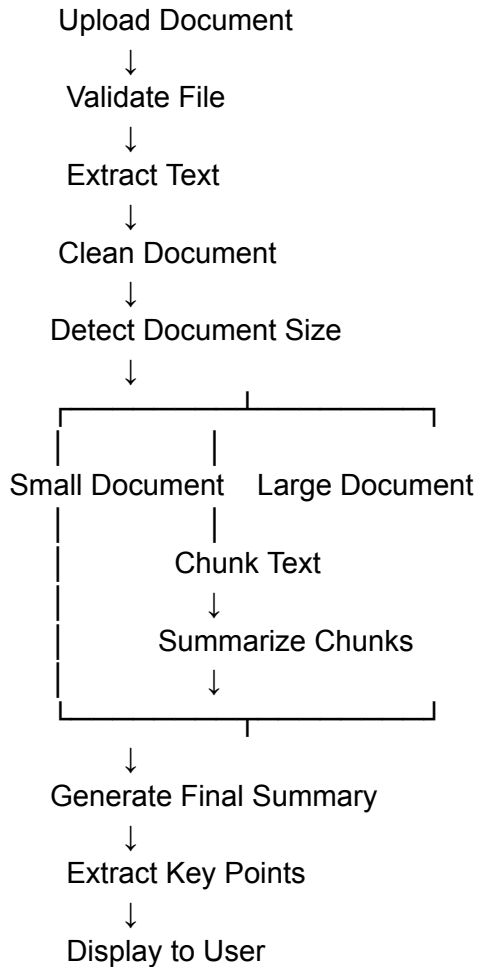


Appendix C – Use Cases

Use Case	Actor	Description
Register	User	Create a new account
Login	User	Authenticate into the system
Upload Document	User	Upload a supported document
Summarize	User	Generate an AI summary
Select Length	User	Choose summary detail level
Extract Key Points	System	Identify important information
Ask Question	User	Ask questions about the document

View History	User	View previous summaries
Download	User	Download generated summary
Manage Users	Admin	Manage registered users

Appendix D – Summary Generation Workflow



Appendix E – Future Enhancements

Future versions may include:

- Multilingual document summarization
- Voice-based document interaction

- Advanced citation extraction
 - Automatic research-paper comparison
 - Research-gap identification
 - AI-generated study notes
 - Personalized summaries
 - OCR support for scanned documents
 - Collaborative document analysis
 - Offline/local LLM support
 - Advanced document analytics
 - Integration with cloud storage platforms
 - Enterprise-level document management
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Appendix F – Project Objective

The overall objective of the **AI-Based Smart Document Summarizer** is to develop an intelligent and user-friendly platform that reduces the time and effort required to understand lengthy documents by automatically extracting and presenting their most important information in a concise and meaningful form.



Decision Table – DocuMind

AI-Based Smart Document Summarizer

Conditions Rules	R1	R2	R3	R4	R5	R6
User logged in?	No	Yes	Yes	Yes	Yes	Yes
Supported document (PDF/DOCX/TXT)?	–	No	Yes	Yes	Yes	Yes
Document successfully processed?	–	–	No	Yes	Yes	Yes
Summary length selected?	–	–	–	Short	Medium	Detailed

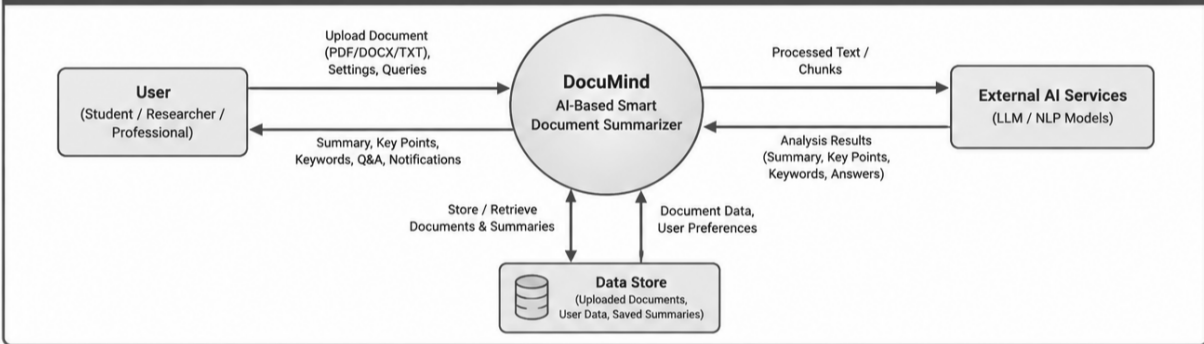
Actions	R1	R2	R3	R4	R5	R6
1. Allow document upload	No	No	Yes	Yes	Yes	Yes
2. Extract & preprocess text	No	No	No	Yes	Yes	Yes
3. Generate AI summary	No	No	No	Yes	Yes	Yes
4. Generate selected summary length	No	No	No	Short	Medium	Detailed
5. Extract key points	No	No	No	Yes	Yes	Yes
6. Detect keywords	No	No	No	Yes	Yes	Yes
7. Enable document Q&A (RAG)	No	No	No	Yes	Yes	Yes
8. Allow save/download	No	No	No	Yes	Yes	Yes



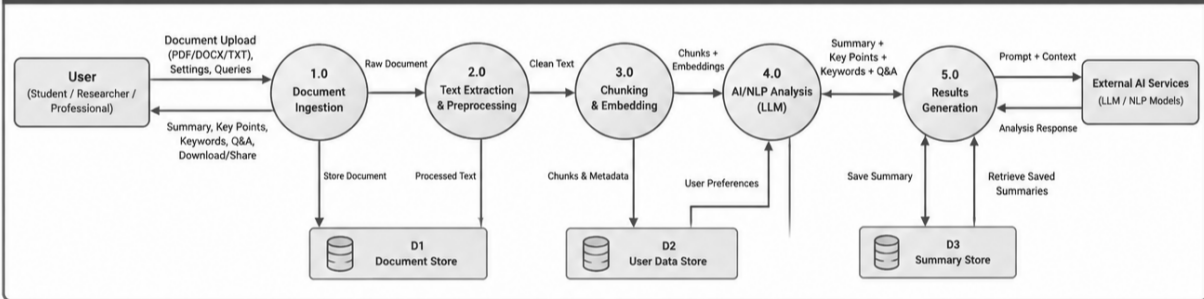
Decision Logic

1. **R1:** User is not logged in → cannot access the document-processing functionality.
2. **R2:** User is logged in but uploads an unsupported file → upload/processing is rejected.
3. **R3:** Supported file is uploaded but text extraction/processing fails → summary cannot be generated.
4. **R4:** Successfully processed document + **Short selected** → short summary and other key information are generated.
5. **R5:** Successfully processed document + **Medium selected** → medium summary and other key information are generated.
6. **R6:** Successfully processed document + **Detailed selected** → detailed summary and other key information are generated.

DFD Level 0 (Context Diagram)



DFD Level 1



DFD Level 2

